

Capacitive Screens

Most popular touchscreen mobiles such as the iPhone and Palm's Pre use this technology in their handsets

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Pioneered multitouch technology at the University of Toronto way back in 1982

And no matter who you are, if you're in a car, having tactile feedback on your dashboard – for changing volume or navigating the GPS – it's going to help you keep your eyes on the road.

How strong is the prototype, and how intensely did you test it?

CH: [Ed. notes: on robustness] We ran a user study – we put several displays in front of users – and I used several of the prototypes for testing input, testing pressure sensitivity, and so on. Overall, they got some decent wear and tear – some hundreds of hours of use. Some of the displays probably got a few hours of use, and there's no noticeable difference. And these are pretty ad-hoc prototypes we made in the lab, with glue and acrylic. So I think a specially engineered model will be pretty robust and airtight.

[Ed. notes: on effectiveness]

We put people at a computer screen, doing something similar to driving tasks – something that really consumes their vision, and we could measure their performance in this game by how accurate they were. So every time they took their eyes off the display to do something else – like press a button – we could measure their performance deficit in that task. Next to the computer monitor, we had another display which showed them which button they needed to press next – “press the upper right button”, “press the big button in the centre”, and so on – and the results, unsurprisingly, were that with a touchscreen, users had to look at it almost every single time, and that degrades their performance. That's a running-over-pedestrians-type deficit. With our buttons and real buttons, they only had to look at the display 20 per cent of the time to find the right button – with touchscreens, it's 90 per cent. So, in a spectrum

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where touchscreens are the worst end and physical buttons are the best, we are really close to the buttons, which is a really positive result. We get all the tactile benefits of the physical buttons, and yet we can display four or five interfaces on a single surface, which is a nice benefit of touchscreens.

Is there any possibility of going beyond the limited number of interfaces?

CH: One of the techniques that we demonstrated is that instead of having just a single layer of latex on top of the display, you actually stack multiple levels of latex, each with its own particular mask – so basically, you're sandwiching latex together, and you can actuate each layer of latex in its own particular way, either one at a time, or a combination at a time. In the end, you can have a large number of different displays. Above five different patterns, this would probably get difficult, but depending on how many different buttons you wanted, you could potentially have a layer per button. The other way to do this is to divide the surface into lots of different chambers. So in the ATM example, there are three different chambers – one for the number pad, one for the buttons on the bottom, and one for the buttons on the side, and so you can individually actuate each one of those independent chambers. So if you wanted to have 100 buttons, the worst case scenario is that you have 100 chambers – probably something you're not going to want to do, but it is an option.

For simple interfaces like

the ATM – five buttons on the screen plus number keypad – you could almost use individual chambers for every button and it wouldn't be terribly expensive.

But if you think about pixels on a computer monitor, they're in the millions. Think of millions of actuators; for the time being, that's not

possible.

Have you tried this at the cellphone scale?

CH: No, but we have ideas on how to do it. Obviously, you can't cut holes in LCD screens on cellphones, so you won't be able to do anything with negative pressure, unless you make it a lot thicker. But we have some good ideas on doing it with some fluidic chambers, and the problem with our big setup is that it requires air pumps, and air pumps are quite sizable – about the size of a computer mouse – but the new system we're developing uses micro-pumps that are hydraulic based, and they're about the size of an American penny, and can be made even smaller. So you can fit a few of these into a mobile device and potentially push up a QWERTY keypad or number keypad right on the surface of the display, which will be really great.

Is it possible with today's technology to go down to that size?

CH: We haven't built anything yet, but I'm pretty convinced it is. Again, you wouldn't be able to do per-pixel, but if you wanted to push up 10 buttons or a number keypad, you can do that on a cellphone.

What further development are you doing on the project right now?

CH: We haven't done a tremendous amount since it was published at the Computer-Human Interface conference about three months ago, because I'm away from CMU and working for Microsoft. We are looking at

miniaturising the displays for use on things like cellphones. But as of right now, we're looking for potential commercial partners to pick up the technology; there's been some interest in the automotive industry.

What problems do you need to address before this technology can make it into the wild?

CH: I think there are quite a few hurdles that still need to be overcome – the first is the robustness of the flexible membrane that's going to be on the top of the display; that's going to have some wear and tear. However, we found that the latex we were using was pretty robust.

Since we're using a rear-projected camera setup, there's a decent amount of throw distance there – so you can't have it as thin as an LCD touchscreen – you're going to need a foot or two of space to project the interface. I think that's probably going to go away, because pico-projectors are getting so small that if you combine them with the right optics, you could make that about a couple of inches [Ed. notes: of projecting distance], maybe less.

It also needs to be robust enough survive in a car environment, which is noisy and air-prone. And since this is air-driven, it may be worthwhile to use some type of fluid. But the problem with any kind of fluid is that leaks are possible.

The technology's pretty good, but it would take... maintenance...to keep your tactile display going. But honestly, the robustness of the prototype we have in the lab is pretty good.

As for cost, they're really very inexpensive, so that's not a hurdle at all.

It may be too early to hope for tactile touchscreens in cellphones, but if they can be cheaply manufactured, it might just mean that this one won't end up on the could-have-been-great-but... pile, and may well be worth watching for. 